TANISHA GAUTAM

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EDUCATION

B.Tech in Computer Science Engineering :NSUT, New Delhi | 2028 | CGPA: 7.49/10
Board Examinations: Class XII (88%) | Class X (91%)

EXPERIENCE

Machine Learning Intern
Defence Research and Development Organisation (DRDO), ISSA | May – July 2026

- Developed a **voxel-based 3D terrain representation framework** using real-world OBJ terrain datasets for accurate modelling of mountainous environments and autonomous UAV navigation.
- Implemented the **A* path planning algorithm** to generate optimal, collision-free navigation paths, improving routing efficiency across complex 3D terrains.
- Optimized **voxel grid resolution** to achieve an effective balance between path accuracy, memory consumption, and computational performance.
- Evaluated **Reinforcement Learning and Neural Network-based approaches** for intelligent navigation and obstacle avoidance, comparing their suitability for autonomous UAV path planning.

PROJECTS

AI-Enabled Early Fault Detection in Industrial Machinery (CPVS)

- Building predictive maintenance models using the **NASA C-MAPSS** turbofan engine database
- Developing **LSTM, CNN-LSTM, and Transformer architectures** for **Remaining Useful Life (RUL)** prediction.
- Creating interactive **Streamlit dashboards** for fault visualization, model monitoring, and explainability.
- Deploying prediction services using **FastAPI**, **Docker**, and **CI/CD pipelines.**

TrackDelivery | Real-time Location Tracking Application

- Developed a real-time location tracking web application using the **MERN** stack, **Socket.io**, and **React-Leaflet.**
- Implemented live location updates with **dynamic distance** and **ETA calculation** using WebSockets.
- Built a responsive UI with **Tailwind CSS** and efficient backend APIs using **Node.js** and **Express**.

Mini Operating System Utility Suite

- Developed a Python-based mini operating system utility suite for **file and directory management.**
- Implemented an **interactive command-line shell** with navigation, compression, backups, and command history.
- Used **multithreading**, **task scheduling**, and **object-oriented programming** for efficient system automation.

House Price Prediction Model

- Built a **regression-based machine learning model** to predict Bengaluru house prices using historical property data.
- Performed **data preprocessing**, **feature engineering,** and **model evaluation** to improve prediction accuracy.
- Integrated the model with a **Flask backend** and responsive **Bootstrap frontend.**

TECHNICAL SKILLS

Languages & Frameworks: Python • JavaScript • React • Angular • C++ • Java • Node.js • Express • Flask • HTML/CSS • Tailwind
ML/Data Tools: TensorFlow • Keras • PyTorch • Pandas • NumPy • Scikit-learn • Matplotlib • Plotly • PyVista • Trimesh
Specialized: 3D Voxel Representation • Pathfinding Algorithms • Deep Learning • OOPs • Docker • CI/CD • WebSockets